

ST PAUL 4 NE, AK, USA

WMO: 703090

Lat: 57.1575N

Lon: 170.2119W

Elev: 6

StdP: 101.25

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 25711

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-18.0	-15.4	-18.6	0.7	-16.0	-16.5	0.9	-14.0	14.6	-6.9	13.4	-5.7	4.5	N/A	0.835

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.5	13.1	11.7	12.3	11.2	11.6	10.7	12.4	13.1	11.8	12.3	11.3	11.8	3.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	8.7	12.5	11.5	8.4	12.1	10.9	8.1	11.5	35.0	13.1	33.6	12.4	32.3	11.8	15.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.5	10.2	DB	-19.1	14.9	4.5	1.5	-22.3	16.0	-24.9	16.8	-27.4	17.6	-30.6	18.7	(4)
(5)			WB	-17.7	13.3	4.6	1.0	-21.0	14.0	-23.7	14.6	-26.3	15.1	-29.6	15.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-4.0	-3.5	-4.8	-1.7	1.9	5.6	8.3	9.5	7.7	4.2	0.9	-1.4	(6)
(7)		DBStd	6.02	4.80	5.35	5.64	3.76	2.68	1.99	1.84	1.55	2.18	2.62	3.00	3.70	(7)
(8)		HDD10.0	2972	434	378	460	351	252	132	59	29	72	180	272	353	(8)
(9)		HDD18.3	5991	693	612	718	601	510	381	311	275	318	438	522	611	(9)
(10)		CDD10.0	22	0	0	0	0	0	0	6	13	3	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.7	6.1	6.2	5.5	5.0	4.0	3.3	3.0	3.3	4.2	4.8	5.5	5.7	(14)
(15)	Precipitation	PrecAvg	592	44	32	31	31	29	33	49	74	71	77	73	56	(15)
(16)		PrecMax	897	127	85	83	82	71	78	109	140	144	158	195	118	(16)
(17)		PrecMin	249	7	0	2	2	3	0	9	4	16	24	22	16	(17)
(18)		PrecStd	119	23	18	20	18	14	19	23	27	30	30	31	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.7	4.6	6.1	9.3	12.7	14.8	14.6	13.0	10.2	7.3	5.5	(19)
(20)			MCWB	3.2	3.7	3.5	4.5	7.2	9.9	13.1	12.6	11.4	9.0	6.4	4.8	(20)
(21)		2%	DB	3.7	3.8	3.6	4.8	7.5	10.6	13.0	13.2	11.9	9.6	6.7	4.7	(21)
(22)			MCWB	2.8	2.9	2.7	3.7	6.0	8.6	11.8	11.9	10.7	8.6	6.0	4.0	(22)
(23)		5%	DB	3.0	3.1	2.7	4.0	6.4	9.5	11.8	12.5	11.2	9.0	6.0	4.0	(23)
(24)			MCWB	2.1	2.2	1.9	3.2	5.2	8.1	11.0	11.5	10.2	8.1	5.3	3.3	(24)
(25)		10%	DB	2.1	2.5	1.6	3.1	5.5	8.6	11.0	11.8	10.6	8.2	5.2	3.3	(25)
(26)			MCWB	1.4	1.7	0.9	2.4	4.5	7.5	10.3	11.1	9.7	7.3	4.3	2.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	4.4	4.1	5.1	7.4	10.6	13.5	13.1	12.3	9.7	6.9	5.4	(27)
(28)			MCDWB	4.1	4.7	4.4	5.8	8.4	12.4	14.5	14.0	12.6	9.9	7.0	5.5	(28)
(29)		2%	WB	3.1	3.5	3.2	4.3	6.4	9.5	12.3	12.6	11.4	9.1	6.4	4.5	(29)
(30)			MCDWB	3.7	3.9	3.7	4.9	7.1	10.7	13.1	13.2	11.8	9.5	6.6	4.8	(30)
(31)		5%	WB	2.6	2.8	2.5	3.7	5.6	8.7	11.5	12.0	10.7	8.6	5.8	3.7	(31)
(32)			MCDWB	3.2	3.3	3.1	4.3	6.3	9.7	12.2	12.5	11.2	9.2	6.1	4.1	(32)
(33)		10%	WB	1.9	2.2	1.6	3.0	5.0	8.0	10.8	11.6	10.2	7.8	5.0	2.9	(33)
(34)			MCDWB	2.5	2.8	2.3	3.7	5.7	8.9	11.3	12.1	10.8	8.5	5.5	3.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.2	4.2	4.7	3.9	4.0	4.2	3.4	3.5	3.9	4.1	3.7	4.2	(35)
(36)			MCDBR	3.3	3.2	4.2	4.0	5.4	5.8	4.6	4.3	4.1	3.2	3.7	4.1	(36)
(37)			MCWBR	3.2	3.2	4.0	3.4	3.8	3.9	3.4	3.4	3.4	3.2	3.9	4.4	(37)
(38)		5% WB	MCDWB	2.9	3.1	4.2	3.5	4.5	5.4	4.6	3.9	3.4	2.9	3.6	4.2	(38)
(39)			MCWBR	3.0	3.3	4.0	3.1	3.5	4.0	3.7	3.2	2.9	3.2	4.0	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.336	0.332	0.401	0.408	0.403	0.392	0.388	0.371	0.338	0.325	0.271	(40)	
(41)		taud	2.211	2.204	2.286	2.096	2.201	2.306	2.401	2.411	2.358	2.318	2.144	2.320	(41)	
(42)		Ebn at Noon	563	692	819	803	827	837	840	817	770	688	498	518	(42)	
(43)		Edh at Noon	52	81	96	136	131	119	107	99	90	71	55	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.40	1.01	2.35	3.54	4.14	4.28	3.69	3.05	2.38	1.39	0.51	0.26	(44)	
(45)		RadStd	0.05	0.11	0.38	0.40	0.30	0.31	0.28	0.31	0.18	0.09	0.06	0.04	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page